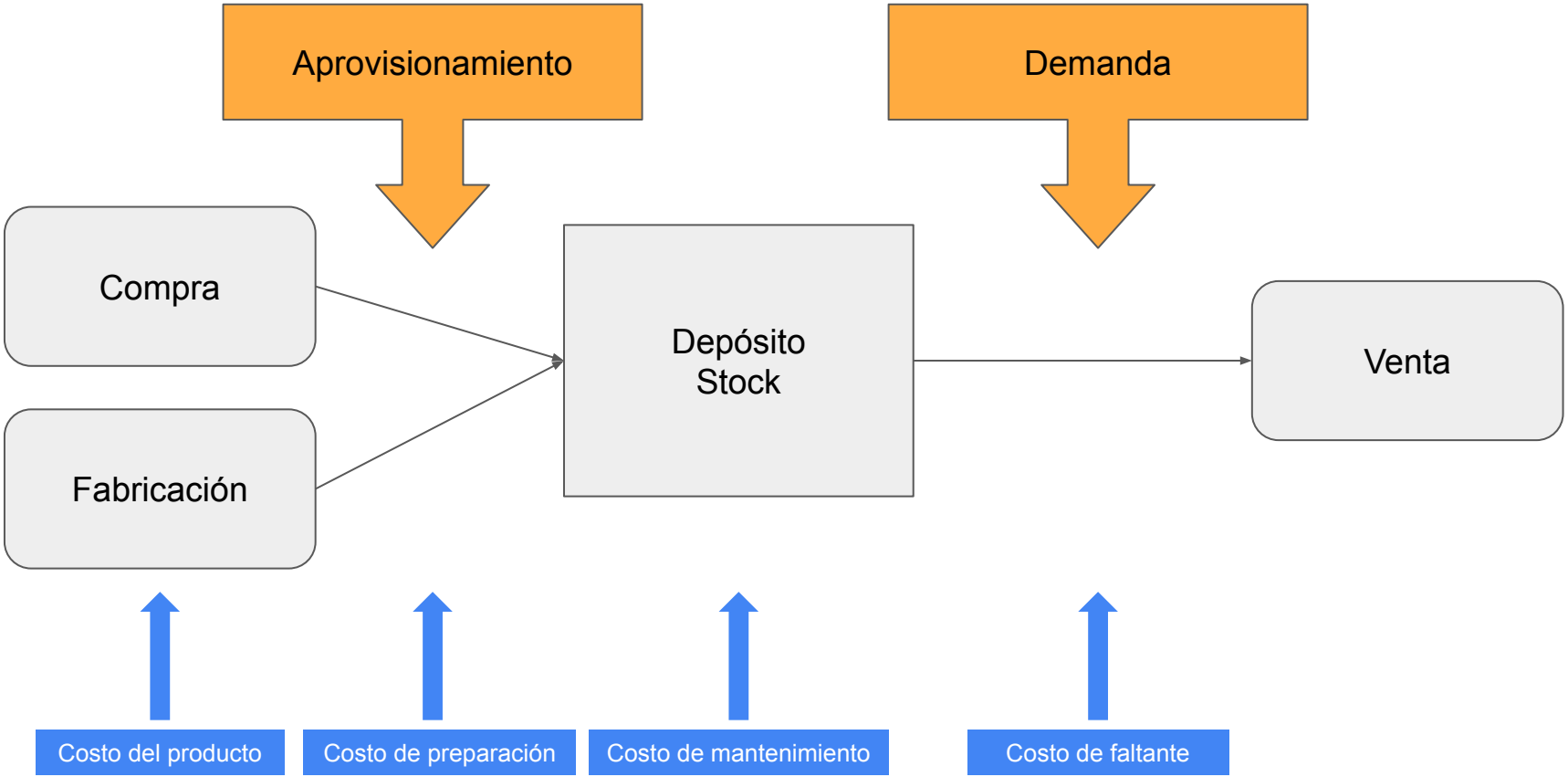
Simulación

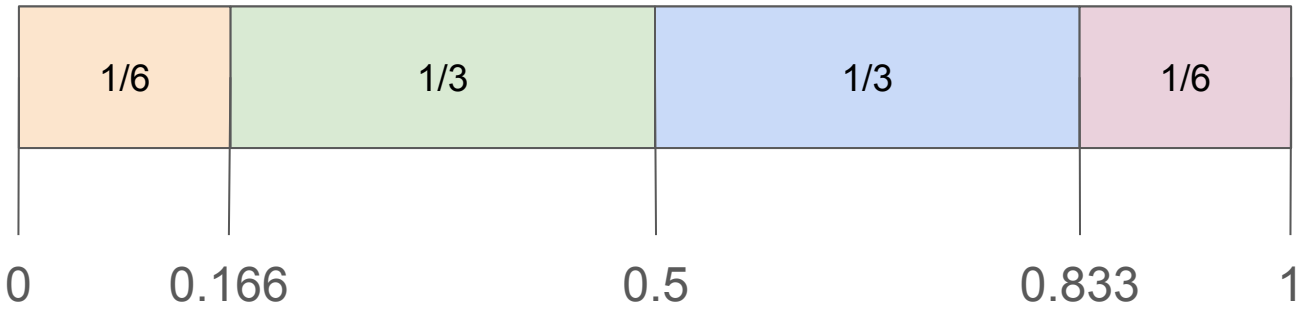
Inventarios

Inventarios: idea conceptual

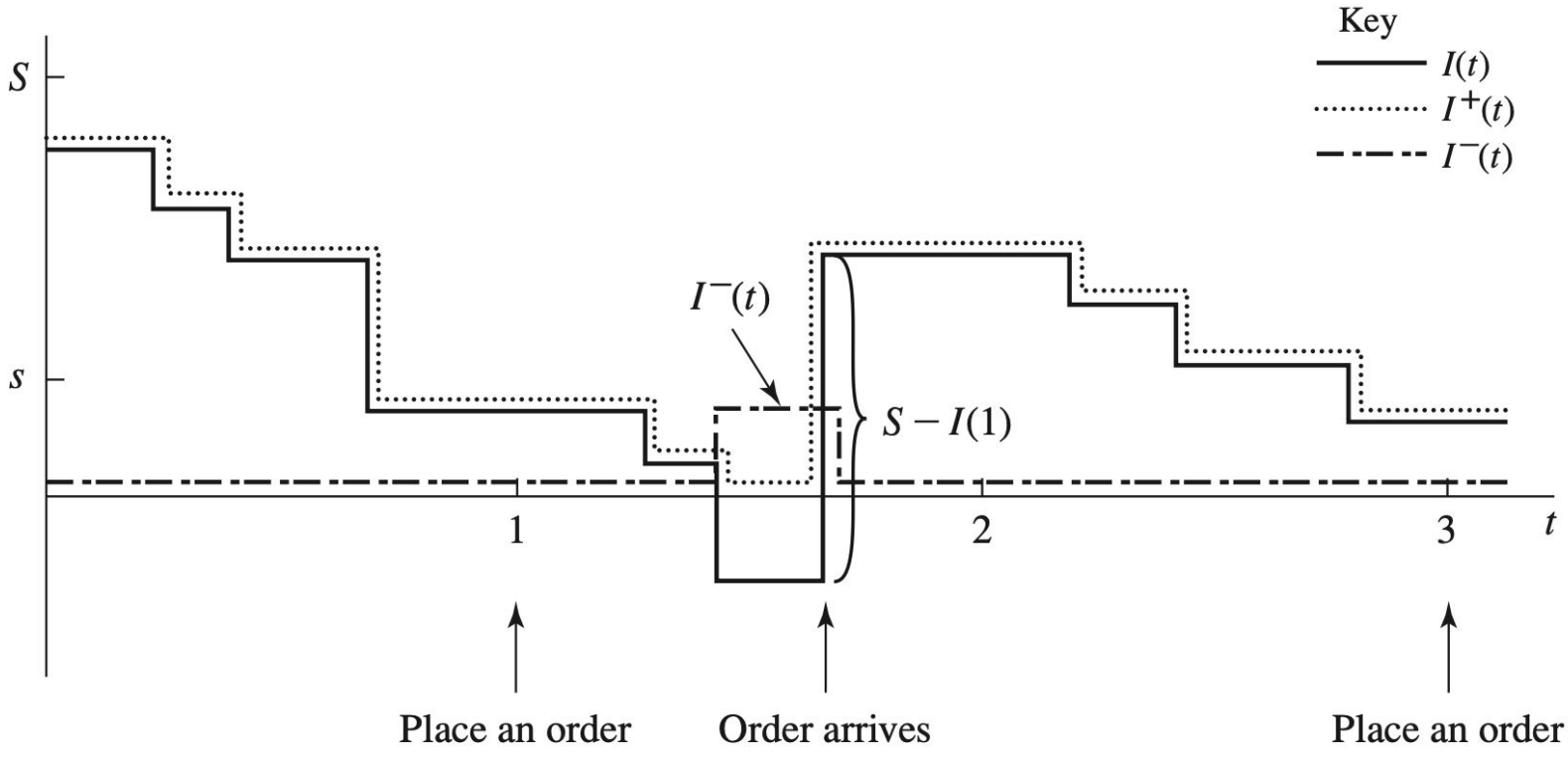


Demanda

- Es aleatoria
- Puede pronosticarse
- Vamos a utilizar distribución empírica



Una realización del nivel de inventarios



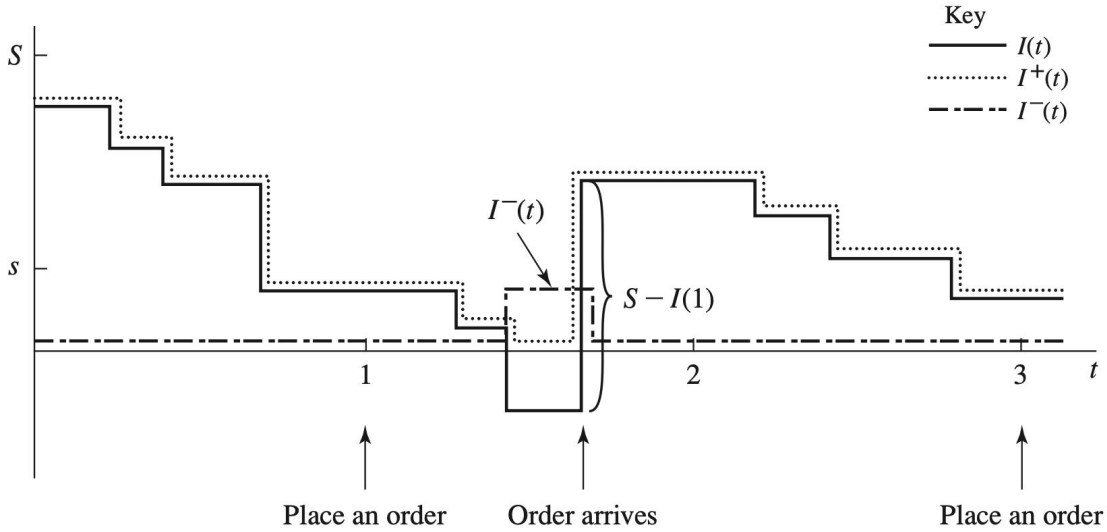
Costos

- Costo unitario del producto (costo incremental) i
- Costo de preparación (setup) k
- Costo unitario de faltante (shortage) pi / p
- Costo unitario de mantenimiento (holding) h
- Costo promedio total

$$C = iZ + k + pI^{-} + hI^{+}$$

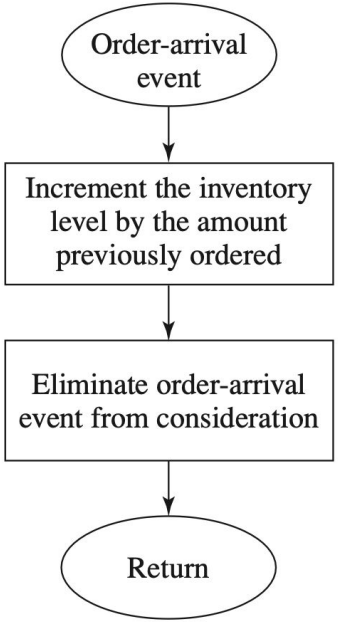
Política de pedidos

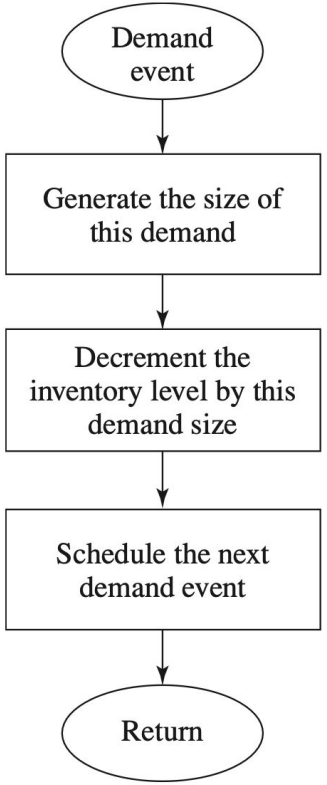
- Revisión periódica: se evalúa el nivel de inventario en períodos de tiempo fijos (cada mes, por ejemplo)
- Política: pedir de acuerdo a la siguiente regla

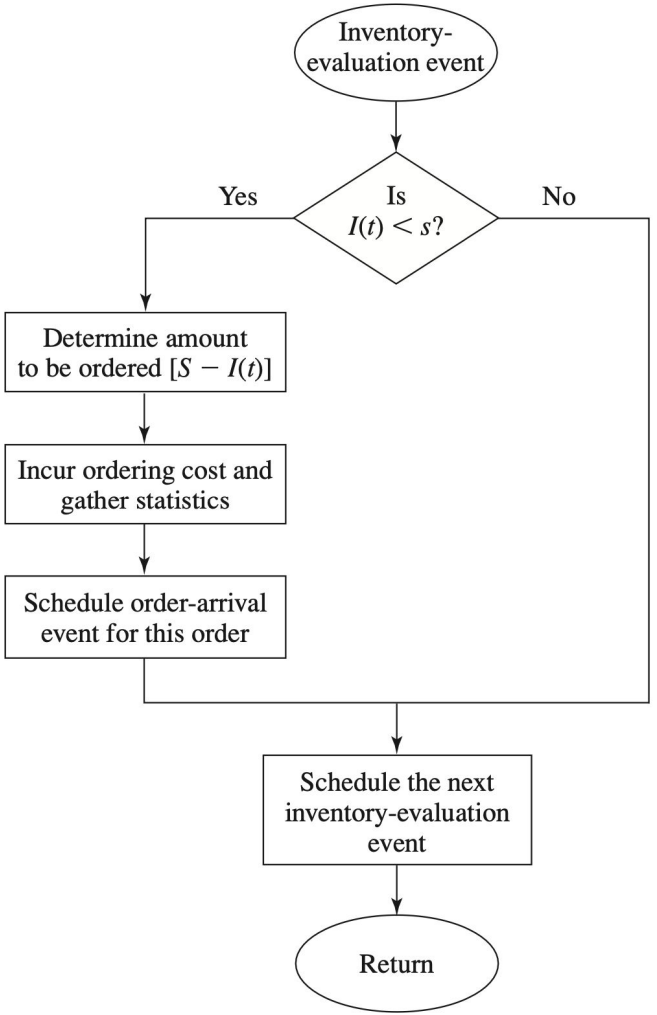


$$Z = \begin{cases} S - I & I < s \\ 0 & I \geq s \end{cases}$$

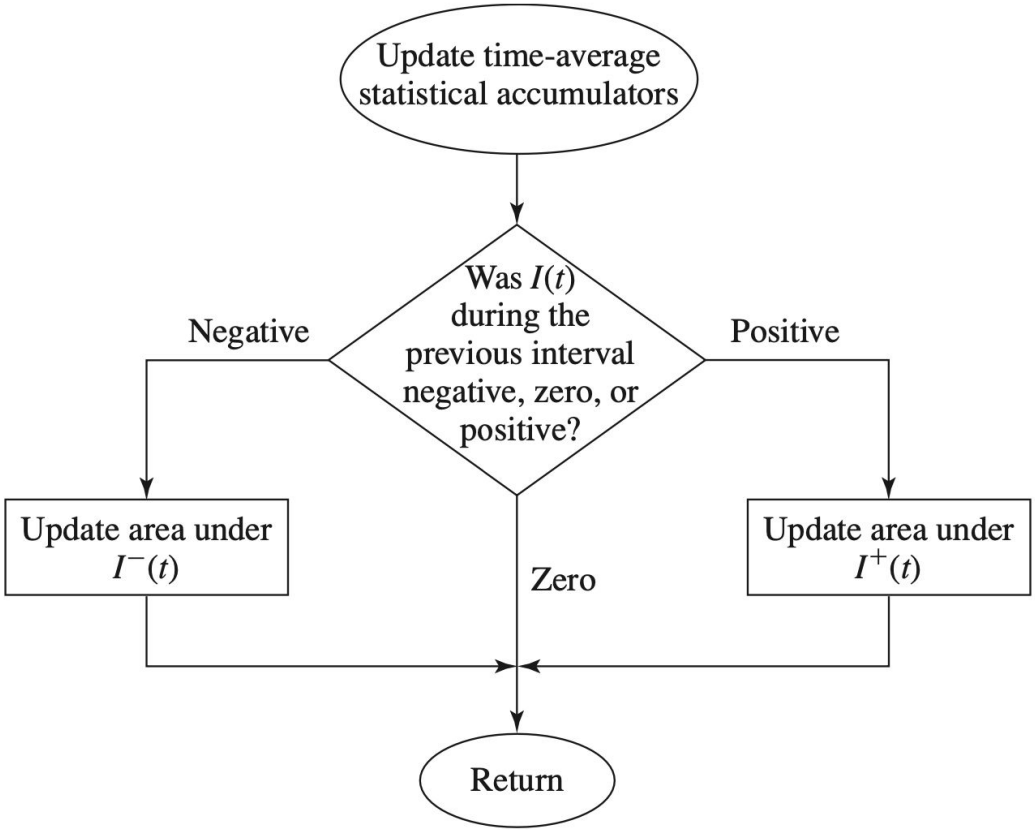
Rutinas


Order-arrival event
Increment the inventory level by the amount previously ordered
Eliminate order-arrival event from consideration
Return


Demand event
Generate the size of this demand
Decrement the inventory level by this demand size
Schedule the next demand event
Return


Inventory-evaluation event
Is $I(t) < s$?
Yes
No
Determine amount to be ordered $[S - I(t)]$
Incur ordering cost and gather statistics
Schedule order-arrival event for this order
Schedule the next inventory-evaluation event
Return

Rutinas


Update time-average statistical accumulators
Was $I(t)$ during the previous interval negative, zero, or positive?
Negative
Positive
Zero
Update area under $I^-(t)$
Update area under $I^+(t)$
Return

Corrida de simulación

Single-product inventory system

Initial inventory level 60 items

Number of demand sizes 4

Distribution function of demand sizes 0.167 0.500 0.833 1.000

Mean interdemand time 0.10 months

Delivery lag range 0.50 to 1.00 months

Length of the simulation 120 months

K = 32.0 i = 3.0 h = 1.0 pi = 5.0

Number of policies 9

Policy	Average total cost	Average ordering cost	Average holding cost	Average shortage cost
(20, 40)	126.61	99.26	9.25	18.10
(20, 60)	122.74	90.52	17.39	14.83
(20, 80)	123.86	87.36	26.24	10.26
(20,100)	125.32	81.37	36.00	7.95
(40, 60)	126.37	98.43	25.99	1.95
(40, 80)	125.46	88.40	35.92	1.14
(40,100)	132.34	84.62	46.42	1.30
(60, 80)	150.02	105.69	44.02	0.31
(60,100)	143.20	89.05	53.91	0.24